TAMIL FONT FIXER

What is the problem with Tamil fonts in Unity Text Mesh Pro?

Unicode Tamil fonts rely on **glyph substitution (GSUB)** and **glyph positioning (GPOS)** features of OpenType. However, Unity does not fully support these features, so some Tamil characters are not rendered correctly in the Unity Editor or TextMesh Pro.

Current solution

The solution is to use fonts that require little or no GSUB and GPOS processing, allowing Unity to render Tamil text correctly. Examples of such fonts include **TSCII** and **TACE16** fonts.

Text encoded in these formats is not compatible with standard Unicode Tamil fonts.

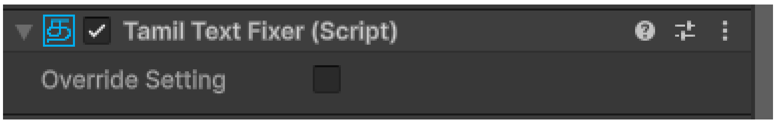
To solve this problem, I developed a built-in Tamil encoder for Unity in C#. It converts Unicode Tamil text into TSCII or TACE16, allowing it to be displayed correctly using TSCII/TACE16 fonts in Unity.

Alternative solution

Use unitext:- https://github.com/LightSideKittens/UniText

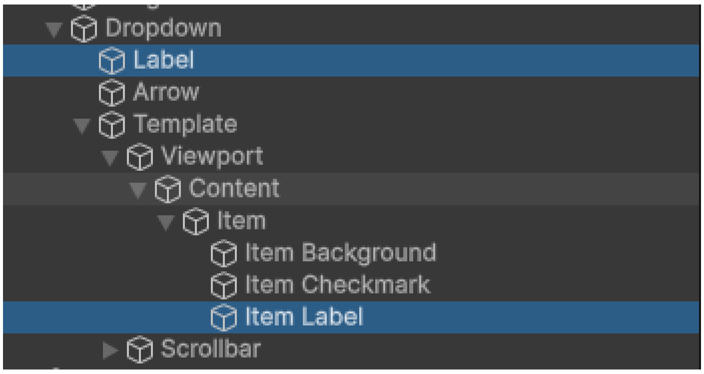
This includes a built-in text rendering engine with support for glyph substitution (GSUB) and glyph positioning (GPOS). However, it does not have plugin support as TextMesh Pro.

USING TAMIL FONT FIXER



Add the Tamil Text Fixer to `TextMeshProUGUI` or `TextMeshPro`. It automatically fixes Tamil text whenever it is assigned, edited, or updated via code in TextMeshPro. Which makes it capable with localization plugins.

Using with Dropdown



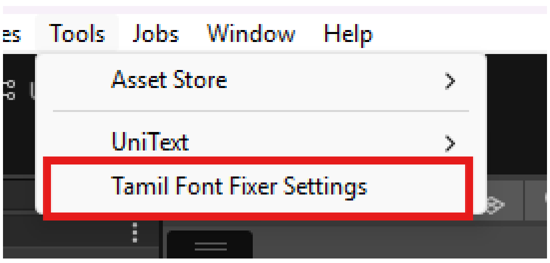
Add the Tamil Text Fixer component to both the Item Label and Head Label. It will work correctly once added.

Using with Input Field

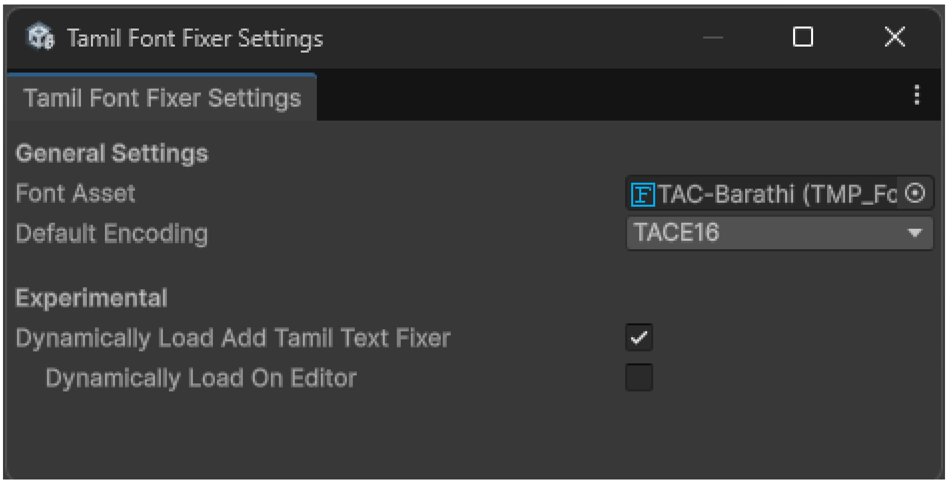
Add the Tamil Text Fixer component to both Placeholder and Text game objects. It will work correctly once added.

TAMIL FONT FIXER SETTING

You can configure the default fonts and encoding from the Tamil Font Fixer Settings. To open the settings, go to Tools → Tamil Font Fixer Settings.



More about settings



Font Asset

Select the default **TextMesh Pro Font Asset** used by the Tamil Text Fixer. This font will be assigned automatically when a compatible font is not already set.

Default Encoding

Select the encoding used by your font. Choosing the correct encoding is essential for proper Tamil text rendering. If the wrong encoding is selected, Tamil characters may not display correctly.

⚠️ **Experimental:**

Dynamically Load Add Tamil Text Fixer

When enabled, the Tamil Text Fixer is automatically added to compatible `TMP_Text` components at runtime. It also assigns the configured font automatically, so no manual setup is required.

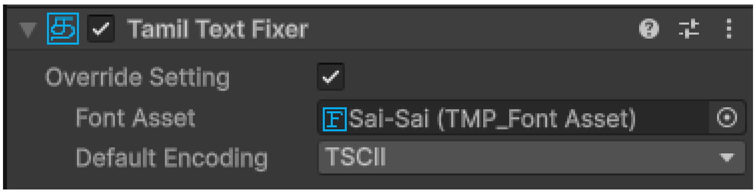
Dynamically Load On Editor

When enabled, the Tamil Text Fixer is also added automatically while working in the Unity Editor, allowing you to preview Tamil text without entering Play Mode.

TAMIL FONT FIXER OVERRIDE OPTION

The **Override** option allows you to use custom font settings for a specific `TamilTextFixer` component instead of the global settings.

When **Override Settings** is enabled, the component ignores the **Tamil Font Fixer Settings** and uses the values specified below.



Override Settings

Enable this option to override the global Tamil Font Fixer settings for the current component.

Font Asset

Select the **TextMesh Pro Font Asset** to use for this component. This font will be applied only to the current text object.

Default Encoding

Select the encoding for the selected font. Make sure it matches the font's encoding; otherwise, Tamil text may not render correctly.

TAMIL FONT FIXER SCRIPTING

The `TamilTextFixer` component exposes the following properties so you can configure its behavior through code.

`bool Deactivated`

Enables or disables the Tamil Text Fixer.

- `true` – The Tamil Text Fixer is disabled and will not process Tamil text.
- `false` – The Tamil Text Fixer is enabled and will process Tamil text normally.

`bool OverrideSetting`

Enables or disables the use of component-specific settings.

- `true` – Uses the component's **Override Font Asset** and **Override Default Encoding**.
- `false` – Uses the global settings configured in **Tools** → **Tamil Font Fixer Settings**.

`TMP_FontAsset OverrideFontAsset`

Sets the TextMesh Pro font asset to use when `OverrideSetting` is enabled.

`TamilFontEncoding OverrideDefaultEncoding`

Sets the font encoding to use when `OverrideSetting` is enabled. Ensure the selected encoding matches the assigned font. otherwise, Tamil text may not render correctly.

TAMIL ENCODER SCRIPTING

The `TamilEncoding` class provides utility methods for converting Tamil text between different font encodings. This is useful when working with legacy Tamil fonts or converting text to match a specific font.

`int TotalTamilCharacter`

Returns the total number of supported Tamil characters used by the encoding system.

```
int count = TamilEncoding.TotalTamilCharacter;
```

`string[] GetCharSet(TamilFontEncoding encoding)`

Returns the character set associated with the specified Tamil font encoding.

Supported Encodings

- `Unicode`
- `TSCII`
- `TACE16`

Returns

- An array containing all characters for the selected encoding.

Example

```
string[] characters = TamilEncoding.GetCharSet(
   TamilFontEncoding.Unicode
);
```

Note: `GetCharSet` throws an exception if an unsupported encoding is specified.

`string Convert(string text, TamilFontEncoding currentFontEncoding, TamilFontEncoding newFontEncoding)`

Converts a string from one Tamil font encoding to another.

Parameters

- `text` – The text to convert.
- `currentFontEncoding` – The encoding of the input text.
- `newFontEncoding` – The target encoding.

Returns

- A string converted to the specified encoding.

Example

```
string converted = TamilEncoding.Convert(
    text,
    TamilFontEncoding.TSCII,
    TamilFontEncoding.Unicode
);
```

`string ConvertFromUnicode(string unicodeCharacters, TamilFontEncoding newFontEncoding)`

Converts Unicode Tamil text to the specified font encoding.

Parameters

- `unicodeCharacters` – The Unicode Tamil text.
- `newFontEncoding` – The target encoding.

Returns

- A string converted from Unicode to the specified encoding.

Example

```
string tsciiText = TamilEncoding.ConvertFromUnicode(
    unicodeText,
    TamilFontEncoding.TSCII
);
```